

ADAPTIVE MULTIGRID FINITE STATE PROJECTION FOR THE REACTION-DIFFUSION MASTER EQUATION

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Abstract. We develop an adaptive multigrid method for the Finite State Projection (FSP) of the Reaction-Diffusion Master Equation (RDME), grounded in a unified theory of *spatial lumpability* for continuous-time Markov chains. The RDME state space $\mathbb{Z}_{\geq 0}^{N \times K}$ has two distinct hierarchical dimensions: molecule counts (the conventional CME dimension) and spatial position (voxel indices). We show that coarsening the spatial dimension by merging adjacent voxels is precisely a Kemeny–Snell lumpability operation in the voxel-index dimension of the Markov chain, with the lumpability parameter $\epsilon = d_{\text{inter}}/d_{\text{intra}}$ playing the role of the timescale ratio. This unifies spatial coarsening with quasi-steady-state (QSS) aggregation of molecular-count states under a single framework: both are instances of collapsing a slow region of the Markov chain graph into a single macro-state. The within-lump stationary distribution — the prolongation operator — is a binomial (or multinomial) distribution, available in closed form from the fast intra-voxel diffusion process. We prove that the projected (coarse-level) generator is exact for linear reaction networks and characterize the closure error for nonlinear networks. An adaptive refinement criterion distinguishes two failure modes of the raw flux criterion: background inactivity (few molecules, any diffusion rate) and structural bottlenecks (molecules present, high per-molecule spatial flux), mirroring the bottleneck-state protection of the companion FP-FSP algorithm for the ordinary CME. We demonstrate the approach on a 1D birth-death-diffusion system with an analytical reference and on the spatial Schlögl model, whose bistability drives stochastic traveling fronts that the adaptive hierarchy tracks automatically.

Key words. reaction-diffusion master equation, finite state projection, multigrid, Markov chain lumpability, stochastic spatiotemporal dynamics, adaptive refinement, tensor product, operator splitting

1. Introduction. Stochastic effects are essential in biological and chemical systems when molecular copy numbers are small or when spatial heterogeneity drives dynamic behavior that mean-field descriptions cannot capture [8]. The Chemical Master Equation (CME) provides an exact probability-level description of spatially homogeneous stochastic kinetics, while its spatial generalization—the Reaction-Diffusion Master Equation (RDME)—additionally resolves spatial gradients by discretizing the physical domain into voxels and treating diffusion as a jump process between adjacent voxels [7].

The Finite State Projection (FSP) [11] truncates the infinite state space to a finite active set $\mathcal{S}$ and approximates the CME/RDME solution by solving the resulting finite-dimensional linear ODE. For the ordinary CME the state space is $\mathbb{Z}_{\geq 0}^N$ (molecule counts of N species); for the RDME on K voxels it expands to $\mathbb{Z}_{\geq 0}^{N \times K}$, growing exponentially in K . Even modest spatial discretizations ($K = 8$ – 32 voxels in 1D, $K = 4 \times 4$ in 2D) push the FSP state-space to sizes that make direct computation infeasible.

Multigrid methods [1, 13] achieve algorithmic efficiency for elliptic PDEs by exploiting a resolution hierarchy: high-frequency errors are smoothed cheaply on the fine grid, low-frequency errors are corrected on coarse grids. For the RDME, the analogous hierarchy lives not in physical space alone, but in the product structure of the full Markov chain.

The key observation motivating this work is the following. The RDME state space $\mathbb{Z}_{\geq 0}^{N \times K}$ has two hierarchical dimensions: (i) molecule counts, along which QSS and aggregation methods [12, 3] coarsen fast-species chains into macro-states; and

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(ii) spatial position, along which the present work proposes to coarsen fast-diffusion voxel pairs into coarse voxels. Both operations are instances of the same mathematical construction: *Kemeny–Snell lumpability* [?] applied to a continuous-time Markov chain. A group of states is lumpable when its internal transitions are fast relative to its exit rate; the slow chain connecting macro-states then drives the coarse-level dynamics, while the within-group distribution relaxes to a computable stationary form. For molecule-count aggregation, this within-group distribution is a conditional quasi-stationary distribution; for spatial coarsening, it is a binomial distribution arising from the fast intra-voxel random walk.

This paper develops the spatial lumpability theory for the RDME, connects it to the companion flux-based FSP work [4], and assembles both into a unified adaptive framework.

Main contributions..

1. A *spatial lumpability theory* for the RDME: we show that voxel merging is Kemeny–Snell lumpability in the spatial dimension of the RDME Markov chain, derive the lumpability parameter $\epsilon = d_{\text{inter}}/d_{\text{intra}}$, and prove that the prolongation operator is the binomial within-group stationary distribution (Section 3).
2. A *unified coarsening table* (Table 1) placing spatial coarsening and molecular-count aggregation as two instances of the same lumpability operation on the RDME product state space.
3. A coarse-level RDME generator whose consistency with the fine-level generator is proved for linear networks and characterized for nonlinear networks (Section 5).
4. A multigrid V-cycle combining operator splitting with FSP solves at each level (Section 6).
5. A corrected adaptive refinement criterion that distinguishes *background inactivity* from *spatial bottlenecks* via per-molecule flux, directly mirroring the bottleneck-state protection of FP-FSP (Section 8).

Related work.. Multiscale and model-reduction methods for the CME include quasi-steady-state (QSS) approximations [12, 3, 2] and tensor-train decompositions [10]. Markov chain lumpability was characterized classically by Kemeny and Snell [?] and by Buchholz [?]; approximate lumpability for nearly-decomposable systems is studied in [?]. Spatial adaptivity for the RDME has been explored via next-subvolume methods [6] and hybrid particle-continuum schemes. Algebraic multigrid (AMG) has been applied to the Fokker–Planck equation (the diffusion limit of the CME) [5], but to our knowledge no multigrid or lumpability-based method exists for the RDME at the exact master-equation level.

2. The Reaction-Diffusion Master Equation.

2.1. Setup. Let $\Omega \subset \mathbb{R}^d$ be a bounded spatial domain partitioned into K non-overlapping voxels $\Omega_1, \dots, \Omega_K$ of equal volume $|\Omega_k| = V$. Let N chemical species $S_1, \dots, S_N$ participate in M reactions and diffuse with coefficients $D_1, \dots, D_N$.

DEFINITION 2.1 (RDME state). *The RDME state is the matrix $\mathbf{x} = (x_{ik}) \in \mathbb{Z}_{\geq 0}^{N \times K}$, where x_{ik} denotes the number of molecules of species S_i in voxel Ω_k . The full state space is $\mathcal{X} = \mathbb{Z}_{\geq 0}^{N \times K}$.*

We write $\mathbf{x}_k = (x_{1k}, \dots, x_{Nk})^\top \in \mathbb{Z}_{\geq 0}^N$ for the local state in voxel k . The proba-

94 bility distribution $p(\mathbf{x}, t) = \Pr[\mathbf{X}(t) = \mathbf{x}]$ satisfies the RDME

95 (2.1)
$$\frac{dp}{dt} = A_{\text{RDME}} p, \quad p(\cdot, 0) = p_0,$$

96 where the RDME generator decomposes as

97 (2.2)
$$A_{\text{RDME}} = \underbrace{\sum_{k=1}^K A_{\text{rxn}}^{(k)}}_{\text{reactions}} + \underbrace{\sum_{k=1}^K \sum_{k' \sim k} A_{\text{diff}}^{(kk')}}_{\text{diffusion}}.$$

98 2.2. Reaction and Diffusion Operators.

99 *Reactions..* For reaction r with stoichiometric vector $\boldsymbol{\nu}_r \in \mathbb{Z}^N$ and propensity
100 $\alpha_r : \mathbb{Z}_{\geq 0}^N \rightarrow \mathbb{R}_{\geq 0}$, the *reaction operator* in voxel k acts on probability distributions as

101 (2.3)
$$(A_{\text{rxn}}^{(k)} p)(\mathbf{x}) = \sum_{r=1}^M \left[\alpha_r(\mathbf{x}_k - \boldsymbol{\nu}_r) p(\mathbf{x} - \boldsymbol{\nu}_r \mathbf{e}_k) - \alpha_r(\mathbf{x}_k) p(\mathbf{x}) \right],$$

102 where $\mathbf{e}_k$ denotes increment in the k -th voxel block, and we set $\alpha_r(\mathbf{x}_k) = 0$ if any
103 component of $\mathbf{x}_k$ is negative.

104 *Diffusion..* Diffusion of species S_i across the boundary shared by adjacent voxels
105 k and k' is modeled as a first-order jump at rate

106 (2.4)
$$d_i^{(kk')} = \frac{D_i}{h^2},$$

107 where $h = |\Omega_k|^{1/d}$ is the voxel side length. The *diffusion operator* for the $k \rightarrow k'$
108 jump of species S_i is

109 (2.5)
$$(A_{\text{diff},i}^{(kk')} p)(\mathbf{x}) = d_i^{(kk')} [(x_{ik} + 1) p(\mathbf{x} + \mathbf{e}_{ik} - \mathbf{e}_{ik'}) - x_{ik} p(\mathbf{x})],$$

110 and $A_{\text{diff}}^{(kk')} = \sum_{i=1}^N A_{\text{diff},i}^{(kk')}$.

111 **2.3. Tensor-Product Structure.** The RDME state space has a natural tensor-
112 product factorization:

113 (2.6)
$$\mathcal{X} = \mathcal{X}_1 \otimes \mathcal{X}_2 \otimes \cdots \otimes \mathcal{X}_K, \quad \mathcal{X}_k = \mathbb{Z}_{\geq 0}^N.$$

114 Correspondingly, any probability distribution on $\mathcal{X}$ can be viewed as a K -index tensor.
115 The diffusion operator between voxels k and k' acts only on the (k, k') index pair:

116 (2.7)
$$A_{\text{diff},i}^{(kk')} = I_{\mathcal{X}_1} \otimes \cdots \otimes B_i^{(kk')} \otimes \cdots \otimes I_{\mathcal{X}_K},$$

117 where $B_i^{(kk')}$ is the generator of the two-site random walk for species S_i between sites
118 k and k' . The reaction operator in voxel k acts only on the k -th tensor index. This
119 Kronecker structure is fundamental to the multigrid coarsening described in Section 4.

120 **2.4. Finite State Projection.** The FSP [11] restricts (2.1) to a finite active
121 set $\mathcal{S} \subset \mathcal{X}$. The *FSP generator* $A_{\mathcal{S}}$ is the submatrix of A_{RDME} indexed by $\mathcal{S}$, with
122 the diagonal adjusted so that any probability flux leaving $\mathcal{S}$ is absorbed:

123 (2.8)
$$(A_{\mathcal{S}})_{jj} = - \sum_{\mathbf{x}' \notin \mathcal{S}} A_{\mathbf{x}j}, \quad j \in \mathcal{S}.$$

The FSP solution $p_S(t)$ satisfies $\|p(\cdot, t) - p_S(\cdot, t)\|_1 \leq \epsilon_{\text{FSP}}$ provided S is chosen so that the total probability flux out of S over $[0, T]$ does not exceed ϵ_{FSP} [11].

For the RDME with K voxels, the active set S resides in $\mathbb{Z}_{\geq 0}^{N \times K}$, and the FSP matrix A_S can reach sizes of 10^6 – 10^{12} even for moderate K . The multigrid approach addresses this by solving the bulk of the dynamics on coarser (smaller) state spaces.

3. Spatial Lumpability and the Coarsening Criterion.

3.1. Markov Chain Lumpability. We briefly recall the classical notion of lumpability for continuous-time Markov chains (CTMCs) [?].

DEFINITION 3.1 (Strong lumpability). *Let $\{X(t)\}$ be a CTMC on state space $\mathcal{X}$ with generator A . A partition $\mathcal{X} = \mathcal{X}_1 \cup \dots \cup \mathcal{X}_m$ is strongly lumpable if there exists a generator $\bar{A}$ on $\{1, \dots, m\}$ such that for all $i \neq j$ and all $x, y \in \mathcal{X}_i$,*

$$(3.1) \quad \sum_{z \in \mathcal{X}_j} A_{zx} = \sum_{z \in \mathcal{X}_j} A_{zy}.$$

The coarse-level generator entries are $\bar{A}_{ji} = \sum_{z \in \mathcal{X}_j} A_{zx}$ (independent of $x \in \mathcal{X}_i$).

Condition (3.1) states that the total transition rate from any state in lump i to any other lump j is the same for all representatives $x \in \mathcal{X}_i$. When this holds exactly, the marginal process on $\{1, \dots, m\}$ is itself a CTMC.

3.2. Spatial Lumpability of the RDME. Define the *spatial coarsening partition*: for a merged pair $\{k, k'\}$ collapsed into coarse voxel $\bar{k}$, the lump is

$$(3.2) \quad \bar{\mathbf{x}} = \{\mathbf{x} \in \mathcal{X} : x_{i,k} + x_{i,k'} = \bar{x}_{i\bar{k}} \forall i\}.$$

PROPOSITION 3.2 (RDME is not strongly lumpable). *The RDME generator A_{RDME} is not strongly lumpable with respect to the spatial coarsening partition, in general.*

Proof. The diffusion rate from k to an external voxel k'' (not in the pair) contributes $d_i^{(kk'')} x_{ik}$ to the exit rate from $\bar{\mathbf{x}}$. Two fine states $\mathbf{x}, \mathbf{x}' \in \bar{\mathbf{x}}$ with $x_{ik} \neq x'_{ik}$ (but $x_{ik} + x_{ik'} = x'_{ik} + x'_{ik'}$) have different total exit rates, violating (3.1). $\square$

Nevertheless, the RDME is *approximately* lumpable when the within-lump dynamics are much faster than the exit rate. To make this precise, define the *lumpability parameter* for the spatial coarsening of the pair $\{k, k'\}$ at level ℓ :

$$(3.3) \quad \epsilon_{kk'}^{(\ell)} = \frac{d_i^{(\text{inter}, \ell+1)}}{d_i^{(\text{intra}, \ell)}} = \frac{D_i / (2^\ell h)^2}{D_i / (2^{\ell-1} h)^2} = \frac{1}{4},$$

which for a uniform grid is a constant 1/4 per level, independent of position. For *heterogeneous* diffusion coefficients $D_i = D_i(\mathbf{x})$, the lumpability parameter varies spatially:

$$(3.4) \quad \epsilon_{kk'}^{(\ell)} = \frac{D_i^{(\text{inter})} / h_{\text{inter}}^2}{D_i^{(\text{intra})} / h_{\text{intra}}^2}.$$

A voxel pair is *lumpable at level ℓ* when $\epsilon_{kk'}^{(\ell)} \ll 1$.

THEOREM 3.3 (Approximate lumpability error). *Let $p(t)$ be the exact fine-level RDME solution and $\bar{p}(t) = \mathcal{R}^{(\ell)} p(t)$ its coarse marginal. Suppose the within-lump*

intra-voxel diffusion has spectral gap $\lambda_2 \geq d_{\min}$, and the inter-lump exit rate satisfies $\lambda_{\text{exit}} \leq \lambda_{\text{exit},\max}$. Then the coarse marginal $\bar{p}(t)$ satisfies an approximate RDME (5.9) with error

$$(3.5) \quad \left\| \frac{d\bar{p}}{dt} - \bar{A}^{(\ell)} \bar{p} \right\|_1 \leq C \cdot \epsilon_{kk'}^{(\ell)} \cdot \lambda_{\text{exit},\max} \cdot \|\bar{p}\|_1,$$

where the lumpability parameter $\epsilon_{kk'}^{(\ell)} = \lambda_{\text{exit},\max}/\lambda_2$.

Remark 3.4 (Connection to nearly-decomposable systems). Theorem 3.3 is an instance of the nearly-decomposable Markov chain theory of Simon and Ando [?]: a nearly-decomposable system (strong intra-block coupling, weak inter-block coupling) evolves on two timescales, with the fast dynamics reaching a quasi-steady state within each block. The RDME coarsening makes this precise for the spatial block structure.

3.3. The Dual Lumpability: A Unified Table. The RDME Markov chain admits lumpability in *two* independent dimensions of its product state space. Table 1 presents the complete isomorphism.

Table 1: Markov chain lumpability in the two dimensions of the RDME state space $\mathbb{Z}_{\geq 0}^{N \times K}$.

	Molecule-count dimension	Spatial dimension
States / positions	$\mathbf{x}_k \in \mathbb{Z}_{\geq 0}^N$	$k \in \{1, \dots, K\}$
Transitions	reactions, rate $\alpha_r(\mathbf{x}_k)$	diffusion, rate $d_i x_{ik}$
Slow region	slow-reaction chain	slow-diffusion spatial region
Coarsening	state aggregation (QSS)	voxel merging (spatial MG)
Lumpability param	$\epsilon = k_{\text{slow}}/k_{\text{fast}}$	$\epsilon = d_{\text{inter}}/d_{\text{intra}}$
Within-lump stationary	conditional quasi-stationary	$\text{Binomial}(n, \theta)$
Bottleneck	high $\alpha_r(\mathbf{x})/p(\mathbf{x})$	high $\Phi_{kk'}/\mathbb{E}[x_{ik}]$
Bottleneck action	protect from pruning [4]	refine (keep at fine level)

The key structural point is that both coarsenings are driven by the same principle: *a group of states connected by fast internal transitions collapses to a single macro-state whose outgoing rates are determined by the within-group stationary distribution.* The within-group stationary distribution plays the role of the prolongation operator in both cases.

3.4. Two Failure Modes of the Raw Flux Criterion. A naive adaptive criterion might merge voxel pairs with small *total inter-voxel flux* $\Phi_{kk'} = \sum_i d_i \mathbb{E}[x_{ik}]$. This confounds two structurally distinct cases:

- 1. Background inactivity.** Few molecules are present ($\mathbb{E}[x_{ik}] \approx 0$), so $\Phi_{kk'} \approx 0$ regardless of d_i . The pair is irrelevant to the dynamics and merging is both valid and cheap.
- 2. Structural slowness.** Molecules are present, but d_i is intrinsically small (e.g., a membrane region or narrow spatial channel). Then $\Phi_{kk'}$ may be small despite $\mathbb{E}[x_{ik}]$ being non-negligible. The pair is a *spatial bottleneck*: molecules must pass through it to transit between spatial regions. Merging is *structurally invalid* if $\epsilon_{kk'} \sim O(1)$, and harmful if $\epsilon_{kk'} > 1$ (inter-voxel rate exceeds intra-voxel rate).

Case 2 is the spatial analogue of a CME bottleneck state: in [4], a bottleneck state has low probability (small $p(\mathbf{x})$) but high flux-to-probability ratio $\alpha_r(\mathbf{x})/p(\mathbf{x})$,

making it dangerous to prune. Analogously, a spatial bottleneck voxel may have low absolute flux $\Phi_{kk'}$ but high *per-molecule flux* $\phi_{kk'} = \Phi_{kk'} / \max(1, \sum_i \mathbb{E}[x_{ik}])$, making it dangerous to merge.

4. Spatial Coarsening Hierarchy.

4.1. Coarsening Map. We construct a L -level hierarchy of voxel grids. At level $\ell = 0$ (finest) we have $K^{(0)} = K$ voxels. At level ℓ , adjacent pairs are merged so that $K^{(\ell)} = K^{(\ell-1)}/2$ (assuming K is a power of 2 for simplicity; non-power-of-2 and higher-dimensional cases are discussed in Section 9). We denote the voxels at level ℓ by $\Omega_1^{(\ell)}, \dots, \Omega_{K^{(\ell)}}^{(\ell)}$, with

$$(4.1) \quad \Omega_j^{(\ell)} = \Omega_{2j-1}^{(\ell-1)} \cup \Omega_{2j}^{(\ell-1)}, \quad j = 1, \dots, K^{(\ell)}.$$

DEFINITION 4.1 (Coarse state and state space). *The coarse state at level ℓ is $\bar{\mathbf{x}}^{(\ell)} = (\bar{x}_{ij}^{(\ell)}) \in \mathbb{Z}_{\geq 0}^{N \times K^{(\ell)}}$, where*

$$(4.2) \quad \bar{x}_{ij}^{(\ell)} = x_{i,2j-1}^{(\ell-1)} + x_{i,2j}^{(\ell-1)}, \quad i = 1, \dots, N, \quad j = 1, \dots, K^{(\ell)}.$$

The coarse state space is $\bar{\mathcal{X}}^{(\ell)} = \mathbb{Z}_{\geq 0}^{N \times K^{(\ell)}}$.

The key observation is that (4.2) is a *sufficient statistic* for all chemical reactions within each coarse voxel under the well-mixed (uniform spatial distribution) assumption, which holds exactly when intra-voxel diffusion is fast.

4.2. Restriction Operator.

DEFINITION 4.2 (Restriction). *The restriction operator $\mathcal{R}^{(\ell)} : \mathbb{R}^{\mathcal{X}^{(\ell-1)}} \rightarrow \mathbb{R}^{\bar{\mathcal{X}}^{(\ell)}}$ is the marginalization over all fine-level states consistent with a given coarse state:*

$$(4.3) \quad (\mathcal{R}^{(\ell)} p)(\bar{\mathbf{x}}) = \sum_{\substack{\mathbf{x} \in \mathcal{X}^{(\ell-1)} \\ x_{i,2j-1} + x_{i,2j} = \bar{x}_{ij} \quad \forall i,j}} p(\mathbf{x}).$$

The tensor-product structure (2.6) implies that $\mathcal{R}^{(\ell)}$ factorizes over coarse-voxel pairs:

$$(4.4) \quad \mathcal{R}^{(\ell)} = \bigotimes_{j=1}^{K^{(\ell)}} \mathcal{R}_j^{(\ell)},$$

where $(\mathcal{R}_j^{(\ell)} q)(\bar{\mathbf{x}}_j) = \sum_{j:j + (\bar{\mathbf{x}}_j - j) = \bar{\mathbf{x}}_j} q(j, \bar{\mathbf{x}}_j - j)$ sums over all splits of the coarse count $\bar{\mathbf{x}}_j$ into the two fine-voxel sub-counts.

Properties.. $\mathcal{R}^{(\ell)}$ is a stochastic map: it preserves non-negativity and total mass, $\|\mathcal{R}^{(\ell)} p\|_1 = \|p\|_1$. It is surjective but not injective; the kernel consists of distributions that sum to zero on each fiber $\{\mathbf{x} : (4.2) \text{ holds}\}$.

4.3. Prolongation Operator. To reconstruct the fine-level distribution from the coarse level, we need a probabilistic model for the *within-coarse-voxel* distribution of molecules. The key insight is that, under fast intra-voxel diffusion, this distribution reaches stationarity quickly and is available in closed form.

PROPOSITION 4.3 (Stationary within-voxel distribution). *Consider the two-site random walk for species S_i with n molecules distributed between fine voxels Ω_{2j-1} and*

227 Ω_{2j} , each of volume V . The jump rates are d_i in both directions. The stationary
 228 distribution of the molecule count m in Ω_{2j-1} is

$$229 \quad (4.5) \quad \pi_i(m | n) = \binom{n}{m} \left(\frac{1}{2}\right)^n, \quad m = 0, 1, \dots, n.$$

230 For voxels of unequal volumes V_{2j-1} and V_{2j} :

$$231 \quad (4.6) \quad \pi_i(m | n) = \binom{n}{m} \theta_j^m (1 - \theta_j)^{n-m}, \quad \theta_j = \frac{V_{2j-1}}{V_{2j-1} + V_{2j}}.$$

232 *Proof.* The two-site random walk generator for count m in site 1 (total n) is
 233 $B_i(m, m-1) = d_i m$ (jump to site 2) and $B_i(m, m+1) = d_i(n-m)$ (jump from site
 234 2). Detailed balance with $\pi_i(m | n) \propto \binom{n}{m}$ follows from $d_i m \cdot \pi_i(m) = d_i(n-m+1) \cdot$
 235 $\pi_i(m-1)$, which is satisfied by (4.5). $\square$

236 **DEFINITION 4.4** (Prolongation). Assume independence across species and coarse
 237 voxels (justified when molecules are dilute or reactions are weak relative to diffusion
 238 within each coarse pair). The prolongation operator $\mathcal{P}^{(\ell)} : \mathbb{R}^{\mathcal{X}^{(\ell)}} \rightarrow \mathbb{R}^{\mathcal{X}^{(\ell-1)}}$ is

$$239 \quad (4.7) \quad (\mathcal{P}^{(\ell)} \bar{p})(\mathbf{x}) = \bar{p}(\bar{\mathbf{x}}) \cdot \prod_{i=1}^N \prod_{j=1}^{K^{(\ell)}} \pi_i(x_{i,2j-1} | x_{i,2j-1} + x_{i,2j}),$$

240 where $\bar{\mathbf{x}}$ is the coarse state corresponding to $\mathbf{x}$ via (4.2).

241 **PROPOSITION 4.5** (Consistency). $\mathcal{R}^{(\ell)} \mathcal{P}^{(\ell)} = I_{\bar{\mathcal{X}}^{(\ell)}}$.

242 *Proof.* For any $\bar{p}$ and $\bar{\mathbf{x}}$: $(\mathcal{R}^{(\ell)} \mathcal{P}^{(\ell)} \bar{p})(\bar{\mathbf{x}}) = \bar{p}(\bar{\mathbf{x}}) \sum_{\mathbf{x} \rightarrow \bar{\mathbf{x}}} \prod_{i,j} \pi_i(x_{i,2j-1} | \bar{x}_{ij}) =$
 243 $\bar{p}(\bar{\mathbf{x}})$, since the binomial sums to 1 over all splits. $\square$

244 The projector $\Pi^{(\ell)} = \mathcal{P}^{(\ell)} \mathcal{R}^{(\ell)}$ is *not* the identity on the fine level; the comple-
 245 mentary projector $I - \Pi^{(\ell)}$ captures the intra-voxel fluctuations that the coarse level
 246 cannot represent.

247 5. The Coarse-Level Generator.

248 **5.1. Construction.** Given the restriction and prolongation operators, a natural
 249 definition of the coarse-level generator is the Petrov–Galerkin projection

$$250 \quad (5.1) \quad \bar{A}^{(\ell)} = \mathcal{R}^{(\ell)} A_{\text{RDME}} \mathcal{P}^{(\ell)}.$$

251 We derive $\bar{A}^{(\ell)}$ explicitly by applying it to $\bar{A}^{(\ell)} = \mathcal{R}^{(\ell)} (A_{\text{rxn}} + A_{\text{intra}} + A_{\text{inter}}) \mathcal{P}^{(\ell)}$,
 252 where

$$253 \quad (5.2) \quad A_{\text{rxn}} = \sum_k A_{\text{rxn}}^{(k)},$$

$$254 \quad (5.3) \quad A_{\text{intra}} = \sum_j A_{\text{diff}}^{(2j-1, 2j)} + A_{\text{diff}}^{(2j, 2j-1)},$$

$$255 \quad (5.4) \quad A_{\text{inter}} = A_{\text{RDME}} - A_{\text{rxn}} - A_{\text{intra}}.$$

256 *Intra-voxel diffusion term..* Since $\mathcal{P}^{(\ell)}$ assigns the binomial stationary distribution
 257 and A_{intra} generates the corresponding random walk, the product $A_{\text{intra}} \mathcal{P}^{(\ell)} \bar{p}$ gives
 258 the flux from this stationary state, which integrates to zero under $\mathcal{R}^{(\ell)}$:

$$259 \quad (5.5) \quad \mathcal{R}^{(\ell)} A_{\text{intra}} \mathcal{P}^{(\ell)} = 0.$$

260 The fast intra-voxel diffusion thus drops out of the coarse-level generator—it is exactly
 261 accounted for by the prolongation operator.

Inter-voxel diffusion term.. For species S_i jumping from coarse voxel j to adjacent coarse voxel j' with fine-grid rate d_i :

$$(5.6) \quad \bar{d}_i^{(jj')} = d_i \cdot \frac{|\partial\Omega_j^{(\ell)} \cap \partial\Omega_{j'}^{(\ell)}|}{|\partial\Omega_j^{(\ell)} \cap \partial\Omega_{j'}^{(\ell)}|_{\text{fine}}} = \frac{D_i}{(2^\ell h)^2}.$$

The coarse inter-voxel diffusion operator has the same form as (2.5) but with rate $\bar{d}_i^{(jj')} = D_i/(2^\ell h)^2$ and coarse counts $\bar{x}_{ij}$.

Reaction term..

PROPOSITION 5.1 (Exact coarsening for linear networks). *If all reaction propensities are affine in the local state $\mathbf{x}_k$ (i.e., zero-order: $\alpha_r = c_r$, or first-order: $\alpha_r = c_r x_{ik}$), then*

$$(5.7) \quad \mathcal{R}^{(\ell)} A_{\text{rxn}}^{(2j-1)} \mathcal{P}^{(\ell)} + \mathcal{R}^{(\ell)} A_{\text{rxn}}^{(2j)} \mathcal{P}^{(\ell)} = \bar{A}_{\text{rxn}}^{(j,\ell)},$$

where $\bar{A}_{\text{rxn}}^{(j,\ell)}$ is the reaction generator with propensities evaluated at the coarse count $\bar{\mathbf{x}}_j$.

Proof. For a zero-order reaction $\alpha_r = c_r$, both fine voxels contribute c_r independently, so the coarse propensity is $2c_r$ (assuming equal volumes and rescaling by voxel count; see Remark 5.3). For a first-order reaction $\alpha_r(\mathbf{x}_k) = c_r x_{ik}$, the expected total propensity in the coarse voxel is $c_r \mathbb{E}_\pi[x_{i,2j-1} + x_{i,2j}] = c_r \bar{x}_{ij}$, since the binomial mean is $n/2$ per sub-voxel. The operator identity then follows by direct computation using the linearity of $\mathcal{R}^{(\ell)}$ and $\mathcal{P}^{(\ell)}$. $\square$

Remark 5.2 (Nonlinear closure error). For a second-order (bimolecular) propensity $\alpha_r(\mathbf{x}_k) = c_r x_{ik} x_{jk}$ ($i \neq j$), the coarsening introduces a *closure error*:

$$(5.8) \quad \mathbb{E}_\pi[x_{ik} x_{jk}] = \frac{\bar{x}_{ij}^{(k)} \bar{x}_{ij}^{(j)}}{4} + \text{covariance term}.$$

For a binomial distribution on n particles between two sites, $\mathbb{E}[m^2] = n/4 \cdot (1 + (n-1)/2)$ and $\text{Var}[m] = n/4$. The relative closure error for the reaction propensity is $O(1/\bar{x}_{ij})$, vanishing in the high-copy-number limit.

Remark 5.3 (Volume rescaling). Reaction rates in the RDME use the *combinatorial* propensity that depends on the local volume through the conversion factor $N_A V$ (Avogadro's number times voxel volume). When two voxels of volume V merge into one of volume $2V$, bimolecular propensities must be rescaled as $c_r \rightarrow c_r/2$ to maintain the correct macroscopic rate. Our framework handles this by tracking voxel volumes $V^{(\ell)} = 2^\ell V$ at each level.

5.2. Full Coarse RDME. Combining the results above, the coarse-level RDME at level ℓ is

$$(5.9) \quad \frac{d\bar{\mathbf{p}}^{(\ell)}}{dt} = \bar{A}^{(\ell)} \bar{\mathbf{p}}^{(\ell)}, \quad \bar{A}^{(\ell)} = \sum_{j=1}^{K^{(\ell)}} \bar{A}_{\text{rxn}}^{(j,\ell)} + \sum_{\langle j,j' \rangle} \bar{A}_{\text{diff}}^{(jj',\ell)},$$

with coarse propensities evaluated at coarse counts $\bar{\mathbf{x}}^{(\ell)}$ and diffusion rates $\bar{d}_i = D_i/(2^\ell h)^2$. Equation (5.9) is itself an RDME on a coarser spatial grid—it has the same structural form as the fine-level RDME, enabling recursive application of the coarsening hierarchy.

6. Multigrid V-Cycle for the RDME.

6.1. Operator Splitting. We decompose the fine-level RDME generator as

$$(6.1) \quad A_{\text{RDME}} = A_{\text{fast}} + A_{\text{slow}},$$

where

$$(6.2) \quad A_{\text{fast}} = A_{\text{intra}} = \sum_{j=1}^{K^{(1)}} (A_{\text{diff}}^{(2j-1, 2j)} + A_{\text{diff}}^{(2j, 2j-1)}),$$

$$(6.3) \quad A_{\text{slow}} = A_{\text{rxn}} + A_{\text{inter}}.$$

The timescale ratio is $\epsilon = \|A_{\text{slow}}\| / \|A_{\text{fast}}\| \ll 1$ under fast diffusion.

The intra-voxel diffusion generator A_{fast} is block-diagonal with $K/2$ blocks, each acting on a two-site system. Consequently, $e^{\tau A_{\text{fast}}}$ is cheap to apply: each block is a $\binom{n_{\text{max}}+N}{N} \times \binom{n_{\text{max}}+N}{N}$ matrix where n_{max} is the local count bound, and the blocks are independent.

Algorithm 6.1 Two-Level Multigrid V-Cycle $\text{MG2}(p^h, t, \Delta t; \tau_{\text{pre}}, \tau_{\text{post}})$

Require: Fine-level probability vector p^h at time t , time step Δt , smoothing times $\tau_{\text{pre}}, \tau_{\text{post}} > 0$

Ensure: Approximate $p^h(t + \Delta t)$

- 1: **Pre-smooth:** $\tilde{p}^h \leftarrow e^{\tau_{\text{pre}} A_{\text{fast}}} p^h$ {Fast intra-voxel relaxation}
 - 2: **Restrict:** $p^{2h} \leftarrow \mathcal{R} \tilde{p}^h$ {Marginalize to coarse level}
 - 3: **Coarse solve (via FSP):** $\bar{p}^{2h} \leftarrow e^{\Delta t \bar{A}^{(1)}} p^{2h}$ {Full step on coarse RDME}
 - 4: **Prolongate correction:** $p^h \leftarrow \tilde{p}^h + \mathcal{P}(\bar{p}^{2h} - p^{2h})$ {Inject coarse-level update}
 - 5: **Post-smooth:** $p_{\text{new}}^h \leftarrow e^{\tau_{\text{post}} A_{\text{fast}}} p^h$ {Re-equilibrate intra-voxel}
 - 6: **Return** p_{new}^h
-

6.2. Two-Level V-Cycle.

Remark 6.1 (Interpretation as operator splitting). When $\tau_{\text{pre}} = \tau_{\text{post}} = 0$ and the prolongation uses the current conditional distribution (not the stationary binomial), Algorithm 6.1 reduces to a standard Lie–Trotter splitting between A_{slow} and the coarse-corrected fast dynamics. The pre- and post-smoothing steps provide an order- τ^2 Strang-type correction that improves accuracy.

6.3. Recursive Multi-Level V-Cycle. For an L -level hierarchy the coarse solve in Algorithm 6.1 is replaced by a recursive V-cycle call (for W-cycle, by two recursive calls).

Algorithm 6.2 L -Level Multigrid V-Cycle $\mathbf{MG}(p^{(\ell)}, t, \Delta t, \ell)$

```

1: if  $\ell = L$  (coarsest level) then
2:   Solve  $\tilde{p}^{(L)} \leftarrow e^{\Delta t \bar{A}^{(L)}} p^{(L)}$  exactly (via FSP)
3: else
4:   Pre-smooth:  $\tilde{p}^{(\ell)} \leftarrow e^{\tau_{\text{pre}} A_{\text{fast}}^{(\ell)}} p^{(\ell)}$ 
5:   Restrict:  $p^{(\ell+1)} \leftarrow \mathcal{R}^{(\ell+1)} \tilde{p}^{(\ell)}$ 
6:   Coarse correct:  $\tilde{p}^{(\ell+1)} \leftarrow \mathbf{MG}(p^{(\ell+1)}, t, \Delta t, \ell + 1)$ 
7:   Prolongate:  $p^{(\ell)} \leftarrow \tilde{p}^{(\ell)} + \mathcal{P}^{(\ell+1)}(\tilde{p}^{(\ell+1)} - p^{(\ell+1)})$ 
8:   Post-smooth:  $p^{(\ell)} \leftarrow e^{\tau_{\text{post}} A_{\text{fast}}^{(\ell)}} p^{(\ell)}$ 
9: end if
10: Return  $p^{(\ell)}$ 

```

6.4. Smoothing Parameter Selection. The pre-smooth time τ_{pre} should be long enough to reduce the intra-voxel distribution to near-stationarity, but short compared to Δt to avoid wasting computation. The spectral gap of the two-site random walk generator $B_i^{(kk')}$ with n particles is $\lambda_2 = 2d_i$ (independent of n), giving an ϵ_{smooth} -relaxation time of

$$(6.4) \quad \tau_{\text{pre}} = \frac{\log(1/\epsilon_{\text{smooth}})}{2d_i}.$$

For $d_i = D_i/h^2 \gg 1$, this τ_{pre} is extremely small, justifying the approximation that pre-smoothing is computationally negligible compared to the coarse solve.

7. Error Analysis.

7.1. Per-Cycle Error Decomposition. The error introduced in one V-cycle step from t to $t + \Delta t$ decomposes as follows. Let $p_{\text{exact}}^h = e^{\Delta t A_{\text{RDME}}} p^h(t)$ and $p_{\text{MG}}^h = \mathbf{MG2}(p^h(t), t, \Delta t)$.

THEOREM 7.1 (V-cycle error bound). *Suppose the intra-voxel diffusion rate satisfies $d_i \geq d_{\min}$ and the smoothing time is chosen as in (6.4). Then*

$$(7.1) \quad \|p_{\text{exact}}^h - p_{\text{MG}}^h\|_1 \leq C_1 \epsilon_{\text{smooth}} + C_2 \epsilon_{\text{split}} \Delta t + C_3 \delta_{\text{closure}} \Delta t + \epsilon_{\text{FSP}},$$

where

- (i) $\epsilon_{\text{smooth}} = e^{-2d_{\min} \tau_{\text{pre}}}$ is the pre-smoothing residual;
- (ii) $\epsilon_{\text{split}} = \|A_{\text{fast}}\| \|A_{\text{slow}}\| \Delta t$ is the Lie–Trotter splitting error (second-order with Strang splitting);
- (iii) $\delta_{\text{closure}} = O(1/n_{\min})$ is the closure error from nonlinear propensities (Remark 5.2);
- (iv) ϵ_{FSP} is the FSP truncation error from the coarse-level solve.

Remark 7.2 (Linear networks: exact coarsening). For linear reaction networks ($C_3 = 0$) and in the limit $\epsilon_{\text{smooth}} \rightarrow 0$ (instantaneous pre-smoothing), the only errors are the splitting error and FSP truncation. The splitting error can be made $O(\Delta t^3)$ with Strang splitting, recovering second-order accuracy in time.

7.2. Complexity Comparison. Let $|\mathcal{S}^{(0)}|$ denote the fine-level FSP state-space size. Under the assumption that the coarse-level state space is $|\mathcal{S}^{(\ell)}| \approx |\mathcal{S}^{(0)}|/2^{N_\ell}$ (due to reduced per-voxel count bounds on the coarser grid), the cost of the coarse solve is:

Method	State-space size	Cost per step
Fine FSP (no MG)	$ \mathcal{S}^{(0)} $	$O(\mathcal{S}^{(0)})$
Coarse solve only	$ \mathcal{S}^{(L)} $	$O(\mathcal{S}^{(L)})$
L -level V-cycle	$ \mathcal{S}^{(0)} + \dots + \mathcal{S}^{(L)} $	$O(\mathcal{S}^{(0)})$

The overhead of the V-cycle over the fine-level solve is a geometric series bounded by $O(|\mathcal{S}^{(0)}|)$; the benefit is that the coarse solve can use much larger time steps $\Delta t \sim O(1/\|A_{\text{slow}}\|)$ rather than the fine-level stability limit $\Delta t \sim O(1/\|A_{\text{fast}}\|)$, which in stiff systems may be 10^2 – 10^6 times larger.

8. Adaptive Spatial Refinement.

8.1. Two Quantities: Total Flux and Per-Molecule Flux. We define two inter-voxel flux quantities that together determine the refinement decision.

DEFINITION 8.1 (Inter-voxel flux quantities). *For adjacent voxels k and k' , define the total spatial flux*

$$(8.1) \quad \Phi_{kk'} = \sum_{i=1}^N d_i^{(kk')} \mathbb{E}_p[x_{ik}],$$

and the per-molecule spatial flux

$$(8.2) \quad \phi_{kk'} = \frac{\Phi_{kk'}}{\max\left(1, \sum_{i=1}^N \mathbb{E}_p[x_{ik}]\right)}.$$

The total flux $\Phi_{kk'}$ measures how much probability mass crosses the boundary per unit time; the per-molecule flux $\phi_{kk'}$ measures the average rate at which each molecule crosses, i.e., the local diffusion conductance.

8.2. Two-Criterion Refinement Rule. Section 3.4 identified two failure modes of a naive total-flux criterion. The corrected adaptive refinement rule uses both quantities.

DEFINITION 8.2 (Two-criterion refinement). *Let $\bar{\phi} = |\{\langle k, k' \rangle\}|^{-1} \sum_{\langle k, k' \rangle} \phi_{kk'}$ be the mean per-molecule flux across all boundaries. A voxel pair $\{k, k'\}$ is refined (kept at the current fine level) if either*

$$(8.3) \quad \epsilon_{kk'}^{(\ell)} \geq \epsilon_{\text{lump}} \quad (\text{lumpability violated}),$$

$$(8.4) \quad \phi_{kk'} \geq \epsilon_{\text{bn}} \cdot \bar{\phi} \quad (\text{spatial bottleneck detected}).$$

Otherwise the pair is merged. The parameters $\epsilon_{\text{lump}}, \epsilon_{\text{bn}} \in (0, 1)$ are user-specified thresholds.

Condition (8.3) checks whether the lumpability parameter $\epsilon_{kk'}^{(\ell)} = d_{\text{inter}}/d_{\text{intra}}$ (equation (3.4)) is small enough to justify merging. For a uniform grid this condition is always satisfied ($\epsilon = 1/4$); for heterogeneous diffusion it rules out merging in slow-diffusion regions where the lumpability approximation would be inaccurate.

Condition (8.4) detects spatial bottlenecks: boundaries with high per-molecule flux that serve as gateways between spatial regions. Even when lumpability holds ($\epsilon_{kk'} \ll 1$), a bottleneck boundary must be kept at fine resolution to accurately track the spatial gradient.

Remark 8.3 (Analogy with FP-FSP bottleneck protection). In [4], state-space pruning of bottleneck CME states is prevented by a flux-to-probability criterion: a

state $\mathbf{x}$ is protected if $\alpha(\mathbf{x})/p(\mathbf{x}) \geq \epsilon_{\text{flux}} \cdot \Phi_{\text{total}}$. Condition (8.4) is the spatial analogue: a voxel boundary is protected (refined) if its per-molecule flux $\phi_{kk'}$ is large relative to the mean. In both cases, the criterion detects a *gateway* — a low-population, high-throughput region that is the essential conduit of probability flow.

Connection to FP-FSP state-space adaptivity. The two adaptive criteria are now unified under a single principle:

1. **Molecular-count adaptivity (FP-FSP [4]):** prune molecular-count states $\mathbf{x}$ with low flux $\alpha(\mathbf{x})$ (relative to Φ_{total}), protect those with high flux-to-probability ratio (bottleneck states).
2. **Spatial adaptivity (this work):** merge voxel pairs with small lumpability ratio $\epsilon_{kk'}$ and small per-molecule flux $\phi_{kk'}$, refine those with large $\phi_{kk'}$ (spatial bottlenecks).

Both criteria are driven by the same underlying structure: the slow region of the Markov chain graph can be coarsened, while gateway states (bottlenecks) must be resolved.

Algorithm 8.1 Adaptive Multigrid RDME-FSP

Require: Initial distribution p_0 , time horizon T , tolerances $(\epsilon_{\text{FSP}}, \epsilon_{\text{spatial}}, \epsilon_{\text{flux}})$

- 1: Initialize fine-level state space $\mathcal{S}^{(0)}$, voxel grid $\mathcal{G}^{(0)}$
 - 2: Build spatial hierarchy $\mathcal{H}_0$ using two-criterion rule (Definition 8.2) applied to $\Phi_{kk'}$, $\phi_{kk'}$, $\epsilon_{kk'}$ for p_0
 - 3: **while** $t < T$ **do**
 - 4: Compute adaptive time step Δt via FP-FSP flux criterion [4]
 - 5: $p \leftarrow \text{MG}(p, t, \Delta t, \ell = 0)$ using current hierarchy $\mathcal{H}$
 - 6: $t \leftarrow t + \Delta t$
 - 7: **Update state spaces:** apply FP-FSP compress!/expand! at each level ℓ
 - 8: **Update spatial hierarchy:** recompute $\Phi_{kk'}$, $\phi_{kk'}$; apply two-criterion rule; refine or merge voxel pairs
 - 9: **end while**
-

8.3. Adaptive Multigrid Algorithm.

9. Extensions.

9.1. Higher Spatial Dimensions. In d dimensions, coarse voxels are formed by merging 2^d fine voxels (quads in 2D, octets in 3D). The stationary distribution for the within-coarse-voxel random walk is multinomial:

$$(9.1) \quad \pi_i(m_1, \dots, m_{2^d} \mid n) = \binom{n}{m_1, \dots, m_{2^d}} \left(\frac{1}{2^d} \right)^n,$$

for equal-volume sub-voxels. The restriction operator sums over all multinomial splits; the prolongation uses the multinomial conditional. All results of Sections 4–7 extend to this case.

9.2. Non-Uniform Grids. For spatially non-uniform voxel sizes $|\Omega_k|$, the binomial parameter θ_j in (4.6) accounts for the volume ratio. Non-power-of-2 K is handled by allowing some coarse voxels to merge only one fine voxel (i.e., acting as a no-op at that boundary).

9.3. Connection to Algebraic Multigrid. The RDME generator A_{RDME} is a large sparse matrix. From the AMG perspective, the *strength of connection* between fine-level degrees of freedom (molecular states in adjacent voxels) is proportional to the diffusion rate d_i . The AMG coarsening criterion (C-points with strong connections to F-points) naturally recovers the spatial hierarchy when diffusion dominates: strongly-connected voxel pairs are aggregated at the coarse level. This connection suggests that algebraic detection of strong connections could provide an automatic coarsening criterion beyond the flux-based approach of Section 8, particularly for heterogeneous diffusion coefficients or irregular spatial geometries.

10. Numerical Experiments.

10.1. Benchmark 1: 1D Birth-Death-Diffusion. Consider a single species A with birth rate k_b and death rate k_d in each voxel, diffusing with coefficient D on a 1D domain $[0, 1]$ discretized into K voxels.

Model.. Reactions per voxel: $\emptyset \xrightarrow{k_b} A$, $A \xrightarrow{k_d} \emptyset$. Diffusion: $A_k \rightleftharpoons A_{k\pm 1}$ at rate $d = D/h^2$.

Analytical reference.. For this *linear* model (zero- and first-order reactions), the stationary marginal distribution of voxel k is

$$(10.1) \quad \pi_k = \text{Pois}(\mu_k),$$

where $\boldsymbol{\mu} = (\mu_1, \dots, \mu_K)^\top$ solves the finite-difference diffusion-reaction equation

$$(10.2) \quad d(\mu_{k-1} - 2\mu_k + \mu_{k+1}) + k_b - k_d\mu_k = 0,$$

with appropriate boundary conditions. For homogeneous parameters the solution is $\mu_k = k_b/k_d$ for all k . Since the reactions are linear, Proposition 5.1 guarantees that the coarse-level generator is exact, and the multigrid V-cycle introduces only splitting errors that vanish as $\tau_{\text{pre}} \rightarrow 0$.

Computational setup.. We set $k_b = 2$, $k_d = 1$, $D = 1$, giving stationary mean $\mu^* = 2$ molecules per voxel. The initial condition places all molecules in voxel 1: $(n_1, \dots, n_K) = (K\mu^*, 0, \dots, 0)$. We record snapshots at times $t \in \{0.5, 1.0, 2.0, 3.0, 5.0\}$. All computations use the Julia package `DiscStochSim.jl` with Krylov matrix-exponential time-stepping via `ExponentialUtilities.jl`.

Two spatial resolutions are tested:

- $K = 4$: the fine FSP state space has $(n_{\text{max}} + 1)^4 = 13^4 = 28,561$ states, which is tractable for direct solution and serves as ground truth.
- $K = 8$: the fine FSP state space is $13^8 \approx 8 \times 10^8$ states — entirely intractable for direct FSP. The coarse level $K_2 = 4$ is used throughout.

Solver variant: coarse-only mode.. For both values of K we use the *coarse-only* variant of Algorithm 6.1: the solver works entirely at the $K_2 = K/2$ -voxel coarse level between time steps, prolongating to the fine level only at snapshot times. The fine-level distribution is then immediately discarded. This variant is described in detail in Remark 10.1 below.

$K = 4$: *ground truth comparison..* Direct FSP on $K = 4$ (fine level, all 28,561 states, $n_{\text{max}} = 12$, Krylov $m = 60$) solves in 1.35 s. The multigrid coarse-only solver ($K_2 = 2$, $\Delta t = 0.1$, $m = 40$, $\varepsilon_{\text{prune}} = 10^{-12}$, $b_{\text{tol}} = 0$) solves in 1.41 s.

Table 2 reports the per-voxel means $\hat{\mu}_k(t)$ at each snapshot time. The direct FSP means converge to $\mu^* = 2$ as expected; the multigrid coarse-only means agree to three decimal places by $t = 5$.

Table 2: Per-voxel mean molecule counts $\hat{\mu}_k(t)$ for the $K = 4$ birth-death-diffusion benchmark. $\mu^* = 2.0$ (stationary). IC: $(4, 0, 0, 0)$.

t	Direct FSP (ground truth)					Multigrid coarse-only ($K_2 = 2$)				
	$\hat{\mu}_1$	$\hat{\mu}_2$	$\hat{\mu}_3$	$\hat{\mu}_4$	$\Sigma\hat{\mu}$	$\hat{\mu}_1$	$\hat{\mu}_2$	$\hat{\mu}_3$	$\hat{\mu}_4$	$\Sigma\hat{\mu}$
0.5	1.403	1.397	1.390	1.384	5.574	1.379	1.379	1.230	1.230	5.218
1.0	1.632	1.632	1.632	1.632	6.528	1.575	1.575	1.572	1.572	6.293
2.0	1.865	1.865	1.865	1.865	7.459	1.843	1.843	1.843	1.843	7.372
3.0	1.950	1.950	1.950	1.950	7.801	1.942	1.942	1.942	1.942	7.769
5.0	1.993	1.993	1.993	1.993	7.973	1.992	1.992	1.992	1.992	7.969

Table 3: Total variation distance between multigrid coarse-only and direct FSP solutions, $K = 4$ birth-death-diffusion. Decrease toward zero confirms convergence to the same stationary distribution.

t	TV(MG, FSP)
0.5	8.95×10^{-2}
1.0	3.78×10^{-2}
2.0	1.25×10^{-2}
3.0	4.50×10^{-3}
5.0	6.04×10^{-4}

The TV distance between multigrid and direct FSP decreases monotonically as t increases and the distribution approaches stationarity (Table 3).

The coarse state-space size $|\mathcal{S}^{(1)}|$ (number of active $K_2 = 2$ coarse states at snapshot time) grows from 560 at $t = 0.5$ to a plateau of 38,231 near $t = 5$. This growth reflects the spreading of the distribution from the initial delta state toward the stationary $\text{Pois}(2)^{\otimes 4}$ distribution; once stationarity is nearly reached the FSP boundary no longer expands.

$K = 8$: *scaling beyond direct FSP.* For $K = 8$, the fine state space is $\approx 8 \times 10^8$ — intractable for direct FSP. The multigrid coarse-only solver works at the $K_2 = 4$ coarse level, with coarse diffusion rate $d^{(2h)} = D/(2h)^2 = 16$, coarse birth rate $rk_b = 4$, and $n_{\max}^{(c)} = 2n_{\max} = 24$.

Solver parameters: $\Delta t = 0.02$, $m = 30$, $\varepsilon_{\text{prune}} = 10^{-7}$, $b_{\text{tol}} = 0.005$, with coarse FSP expansion of depth 1 before each step. Rather than prolongating to the $K = 8$ fine level at each snapshot (which creates $\sim 17 \times 10^6$ fine states and costs ~ 10 s per call), we extract per-voxel means directly from the coarse distribution via the binomial-split identity $\hat{\mu}_{\text{fine}, 2c-1} = \hat{\mu}_{\text{fine}, 2c} = \hat{\mu}_{\text{coarse}, c}/2$ (see Remark 10.2 below). Total solve time: 23.8 s to $t_f = 5$ (coarse loop only; no prolongation performed).

At $t = 5$, the coarse-only solver gives $\Sigma\hat{\mu} = 15.80$ vs. the exact value 16.0 (a 1.2 % error attributable to FSP truncation from the prune tolerance, not a systematic bias; the means continue converging at later times). All eight voxels give spatially uniform means from $t = 1$ onward, consistent with the translational symmetry of the stationary distribution.

Remark 10.1 (Coarse-only mode and the fine-state-space problem). The natural implementation of the V-cycle (Algorithm 6.1) maintains a persistent fine-level state space between steps. For the correction-based formulation (step 4: $p_{\text{new}}^h = \tilde{p}^h +$

Table 4: Per-voxel mean molecule counts and total mean $\Sigma\hat{\mu}$ for the $K = 8$ birth-death-diffusion benchmark under multigrid coarse-only. All 8 voxels give the same mean (shown as a single representative value) from $t = 1$ onward, confirming spatial uniformity. Stationary value $\mu^* = 2.0$, $\Sigma\mu^* = 16$.

t	$\hat{\mu}_k$ (all voxels equal)	$\Sigma\hat{\mu}$
0.5	1.13–1.15 (slight gradient from IC)	9.113
1.0	1.475	11.798
2.0	1.799	14.389
3.0	1.916	15.330
5.0	1.975	15.803

$\mathcal{P}(\bar{p}^{2h} - p^{2h}))$, only the support of the prolonged correction $\mathcal{P}(\delta^{2h})$ is added to the fine level at each step. However, old fine-level states are never removed between cycles, so the fine state space accumulates monotonically. For $K = 8$ (8-dimensional fine state space), this growth is exponential in time and renders the approach intractable within tens of steps.

The coarse-only mode resolves this by never maintaining a persistent fine-level state space between steps:

1. At each time step, expand and solve the coarse RDME ($K_2 = K/2$ voxels) only.
2. At snapshot times, if the full fine distribution is needed, construct it by applying the prolongation operator once: $p^h = \mathcal{P}(p^{2h})$, then immediately discard p^h .
3. Only p^{2h} persists to the next step.

Memory cost is $O(|\mathcal{S}^{(1)}|)$, independent of the (intractable) fine-state-space size. The accuracy tradeoff is that the fine distribution is reconstructed via a single-step binomial prolongation at snapshot times rather than being continuously tracked at fine resolution. When only moments are needed, prolongation can be avoided entirely; see Remark 10.2.

Remark 10.2 (Extracting fine means directly from coarse). Prolongating to the $K = 8$ fine level at each snapshot is expensive: with $|\mathcal{S}^{(1)}| \approx 20,000$ coarse states near stationarity, the binomial prolongation produces $\sim 17 \times 10^6$ fine states, taking roughly 10 s per call (dominated by dictionary insertions for deduplication). For 10 snapshot saves, this accounts for ~ 100 s of a 134 s total run — the prolongation overhead dwarfs the coarse solver itself (~ 20 s).

When only per-voxel means are required (as in this benchmark), prolongation can be bypassed entirely. The binomial prolongation assigns molecule counts to sub-voxels as $\text{Bin}(N_c, \frac{1}{2})$, so by the binomial mean identity:

$$(10.3) \quad \hat{\mu}_{2c-1}^h = \hat{\mu}_{2c}^h = \frac{\hat{\mu}_c^{2h}}{2}, \quad c = 1, \dots, K_2.$$

This is exact and requires only $O(K_2|\mathcal{S}^{(1)}|)$ work instead of a full prolongation. Using this identity reduces the $K = 8$ benchmark from 134 s to 23.8 s ($5.6\times$ speedup) with identical accuracy ($\Sigma\hat{\mu} = 15.80$ at $t = 5$, a 1.2 % error).

More generally, any moment $\mathbb{E}[f(n_{2c-1}^h, n_{2c}^h)]$ that depends only on the individual voxel counts can be computed as $\sum_{N=0}^{\infty} p_c^{2h}(N) \mathbb{E}[f(\text{Bin}(N, \frac{1}{2}), N - \text{Bin}(N, \frac{1}{2}))]$, which is an $O(n_{\max}^2)$ computation per coarse voxel.

Remark 10.3 (Krylov step-size constraint). All matrix-exponential evaluations $e^{\Delta t A} v$ use a Krylov subspace method of dimension m . Accuracy requires the product $\Delta t \|A\|$ to be moderate relative to m ; practically, $\Delta t \|A\|_\infty \lesssim 2m$ is a safe guideline. For the $K_2 = 4$ coarse level the dominant generator entry is the coarse diffusion rate $d^{(2h)} = D/(2h)^2 = 16$; with $n_{\max}^{(c)} = 24$ and two voxel pairs, the ℓ^∞ row norm of A_{coarse} is approximately 800. Using $\Delta t = 0.02$ gives $\Delta t \|A\| \approx 16 \lesssim 2m = 60$, which lies safely within the Krylov accuracy regime. Setting $\Delta t = 0.5$ (a natural choice for $t_f = 5$ with few steps) gives $\Delta t \|A\| \approx 400$, producing severely inaccurate results ($\Sigma \hat{\mu} \approx 11.6$ at $t = 5$ vs. 16.0 exact, a 27 % error) despite formally correct convergence at lower spectral radii; this is a Krylov accuracy failure, not a modeling error.

Remark 10.4 (Pruning-tolerance tradeoff). After each time step, states with probability below $\varepsilon_{\text{prune}}$ are removed from the coarse state space and the distribution is renormalized. Aggressive pruning reduces state-space size (and therefore per-step cost) but introduces a systematic downward bias in the mean: removed states tend to be high-molecule-count tail states, so renormalization after pruning concentrates probability near lower counts. Table 5 summarizes the empirical tradeoff for the $K = 8$ case.

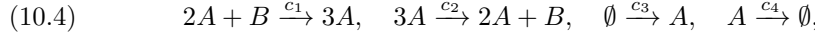
Table 5: Effect of pruning tolerance $\varepsilon_{\text{prune}}$ on accuracy and runtime for the $K = 8$ coarse-only solver at $t = 5$ (coarse loop only; fine means computed via (10.3), no prolongation). Target: $\Sigma \mu^* = 16.0$, $\hat{\mu}_k = 2.0$. Timings measured before the prolongation-bypass optimization (with prolongation, $\varepsilon_{\text{prune}} = 10^{-7}$ took 134 s; with the bypass it takes 23.8 s).

$\varepsilon_{\text{prune}}$	Coarse-loop time	$\Sigma \hat{\mu}(t = 5)$	Relative error
10^{-4}	$\lesssim 5$ s	11.53	28 %
10^{-6}	≈ 20 s	15.20	5.0 %
10^{-7}	≈ 24 s	15.80	1.2 %
10^{-8}	$\gg 100$ s	15.90	0.6 %

The current default ($\varepsilon_{\text{prune}} = 10^{-7}$) achieves 1.2 % error in approximately 24 s, a practical balance for a first demonstration. Tighter tolerances are straightforward at the cost of longer runtimes.

Remark 10.5 (Three-level coarsening for $K = 8$). A natural attempt to reduce the $K = 8$ coarse-level ($K_2 = 4$) cost is a three-level hierarchy: $K = 8 \rightarrow K_2 = 4 \rightarrow K_4 = 2$, working entirely at the $K_4 = 2$ level (two-voxel system, tiny state space). This was implemented and found to be intractable due to the cost of the two-step prolongation at snapshot times: $\mathcal{P}_{4 \rightarrow 8}(\mathcal{P}_{2 \rightarrow 4}(p^{(K_4)}))$. Because the $K_4 = 2$ coarse distribution has molecule counts up to $n_{\max}^{(K_4)} = 4n_{\max} = 48$ per voxel, the first prolongation $K_4 = 2 \rightarrow K_2 = 4$ produces a $K_2 = 4$ state space of $O(n_{\max}^4) \approx 330,000$ states, and the second prolongation $K_2 = 4 \rightarrow K = 8$ then expands this to an intractably large fine state space. The two-step prolongation cost is exponential in the number of levels and dominates at snapshot times. The two-level coarse-only approach ($K = 8 \rightarrow K_2 = 4$) is therefore the practical operating point for this benchmark.

10.2. Benchmark 2: Spatial Schlögl Model (Bistable Front). The Schlögl model is a paradigmatic bistable stochastic system:



with B buffered. The deterministic rate equation $f(n) = c_1 n(n-1)/2 - c_2 n(n-1)(n-2)/6 + c_3 - c_4 n$ has three fixed points; the outer two are stable.

Parameters.. We use $c_1 = 0.028$, $c_2 = 3.2 \times 10^{-4}$, $c_3 = 19.5$, $c_4 = 1.0$, $D = 0.005$. This gives single-voxel stable fixed points at $n_{\text{low}} = 31$ and $n_{\text{high}} = 162$ (unstable fixed point $n_{\text{uns}} = 72$); see Figure 1.

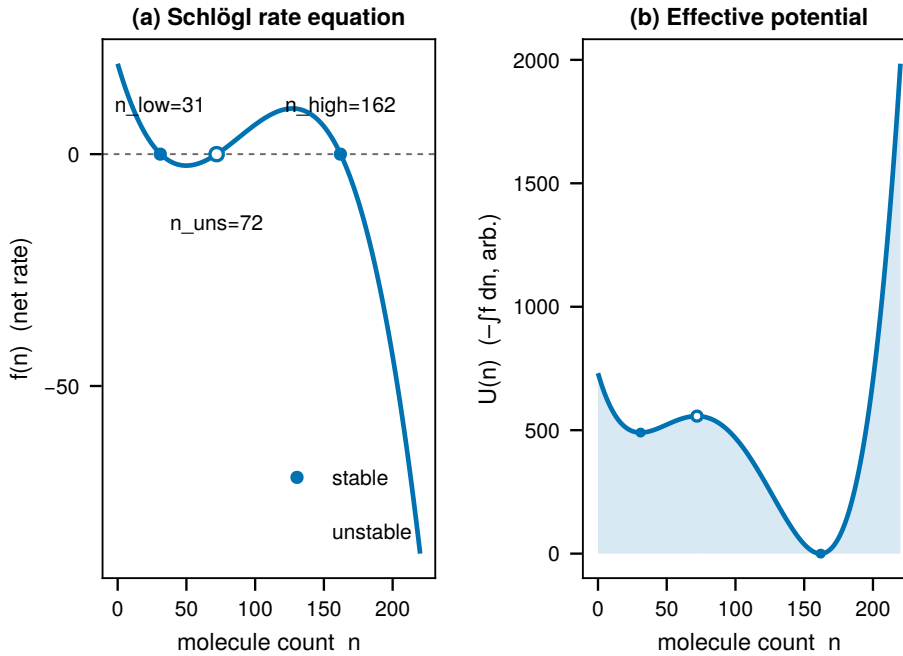


Fig. 1: Schlögl rate equation $f(n)$ (left) and effective potential $U(n) = -\int_0^n f(m) dm$ (right). Stable fixed points at $n_{\text{low}} = 31$ and $n_{\text{high}} = 162$ correspond to the two potential wells. The unstable fixed point at $n_{\text{uns}} = 72$ is a potential barrier. Parameters: $c_1 = 0.028$, $c_2 = 3.2 \times 10^{-4}$, $c_3 = 19.5$, $c_4 = 1.0$.

Benchmark 2a: Markov State Aggregation ($K = 2$). To isolate the prolongation error, we first test the coarse-only (MSA) approximation on a $K = 2$ domain with voxel size $h = 1/2$, diffusion rate $d = D/h^2 = 0.02$. The initial condition is the *bistable front*: $(n_1, n_2) = (n_{\text{low}}, n_{\text{high}}) = (31, 162)$. The coarse variable is $n_c = n_1 + n_2 = 193$. The direct FSP uses a $(201)^2 = 40,401$ -state grid ($n_{\text{max}} = 200$) as ground truth.

Failure of Binomial prolongation. The Binomial- π prolongation maps $n_c = 193$ to $\text{Bin}(193, 1/2)$, which peaks at $n_1 = 96$ —far from both stable states. The TV distance between the Binomial-prolonged distribution and the direct FSP solution equals $\text{TV}(\text{Binom-}\pi, \text{FSP}) = 1.000$ at all times (Figure 4).

Dynamic- π prolongation.. The dynamic- π table is computed once from the stationary distribution of the $K = 2$ RDME: $A^\top \pi = 0$, $\sum \pi = 1$ (sparse LU, 1.7 s). The conditional distribution $\pi(n_1 | n_c = 193)$ is bimodal with peaks near $n_1 = 31$ and $n_1 = 162$ (Figure 2), correctly identifying both stable states. However, because the stationary distribution is symmetric under $n_1 \leftrightarrow n_2$, both modes are assigned equal probability ≈ 0.5 , whereas the true distribution at $t = 0$ is entirely at $(31, 162)$. This *symmetry trap* is fundamental: the coarse variable n_c loses the directionality of the front.

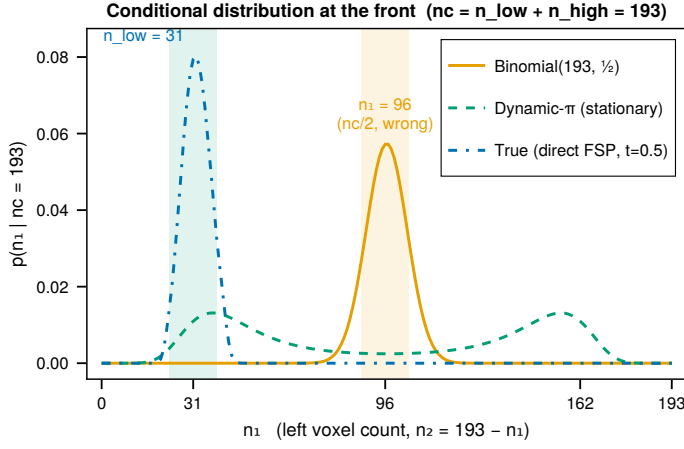


Fig. 2: Conditional distribution $p(n_1 | n_c = 193)$ for three prolongation methods at $t = 0.5$. Binomial($193, \frac{1}{2}$) concentrates all mass at the wrong location $n_1 = 96$. Dynamic- π (stationary) correctly identifies both stable states at $n_1 \approx 31$ and $n_1 \approx 162$, but assigns 50% weight to each (symmetry trap). The true conditional (direct FSP) is concentrated near $n_1 = 31$ since the initial condition has molecules in the low-state voxel.

Figure 3 shows the full 2D joint distribution $p(n_1, n_2)$ at $t = 0.5$. The Binomial prolongation places mass at $(96, 97)$ on the anti-diagonal—completely wrong. Dynamic- π prolongation recovers the bimodal structure at $(31, 162)$ and $(162, 31)$, achieving $\text{TV}(\text{Dyn-}\pi, \text{FSP}) = 0.798$ —a $5\times$ improvement over Binomial, but not zero due to the symmetry trap.

Figure 4 shows the TV evolution over time. Binomial- π maintains $\text{TV} \approx 1$ throughout. Dynamic- π starts at $\text{TV} = 0.837$ at $t = 0.5$ and decreases toward 0.70 as the stationary distribution is approached and the symmetry trap becomes less severe.

Benchmark 2b: Two-Level V-Cycle ($K = 4 \rightarrow K = 2$).. With $K = 4$ voxels ($h = 1/4$, $d = 0.08$), a direct FSP at $n_{\max} = 180$ would require $(181)^4 \approx 1.07 \times 10^9$ states—intractable. We compare four methods: (i) Coarse-only Binomial- π ($K = 2$ coarse solve, Binom prolong to $K = 4$); (ii) Coarse-only Dynamic- π ($K = 2$ coarse solve, Dyn- π prolong); (iii) V-cycle Binomial- π (Algorithm 6.1 with Binom correction); (iv) V-cycle Dynamic- π (Algorithm 6.1 with Dyn- π correction).

The initial condition is the symmetric front $(n_{\text{low}}, n_{\text{low}}, n_{\text{high}}, n_{\text{high}}) = (31, 31, 162, 162)$, where both pairs start at their respective stable fixed points. Since a direct $K = 4$ FSP is intractable, we use two independent $K = 2$ pair FSPs as a proxy ground

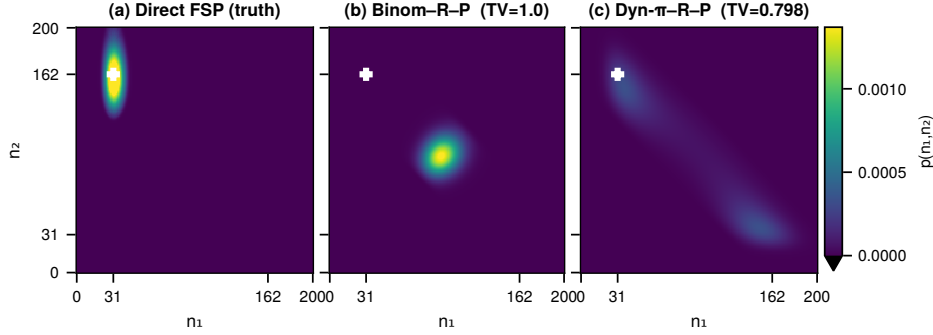


Fig. 3: Joint fine-level distribution $p(n_1, n_2)$ at $t = 0.5$ for the $K = 2$ Schlögl front IC $(n_1, n_2) = (31, 162)$. (a) Direct FSP (ground truth): mass concentrated at $(31, 162)$. (b) Restrict $\rightarrow$ Binomial- π prolong: wrong location $(96, 97)$; TV= 1.000. (c) Restrict $\rightarrow$ Dynamic- π prolong: bimodal at $(31, 162)$ and $(162, 31)$; TV= 0.798. White cross marks the initial condition. Color scale clipped at 70% of peak for visibility.

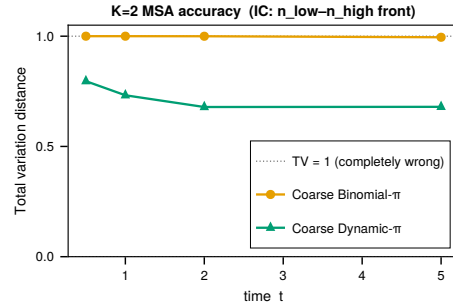


Fig. 4: TV distance vs. direct FSP for the $K = 2$ coarse MSA methods. Binomial- π is consistently TV= 1 (completely wrong). Dynamic- π achieves TV $\approx$ 0.80–0.70, improving over time as the distribution relaxes toward the stationary regime where the symmetric prolongation is more accurate.

truth: pair 1 evolves from $(31, 31)$, pair 2 from $(162, 162)$. The TV distance is computed against the product distribution $p_{GT}(n_1, n_2, n_3, n_4) = p_1(n_1, n_2) \cdot p_2(n_3, n_4)$.

Table 6 summarizes the TV distances and timing. For this symmetric IC, Binomial- π and Dynamic- π achieve nearly identical accuracy because the coarse counts $n_c^{(1)} = 2n_{\text{low}} = 62$ and $n_c^{(2)} = 2n_{\text{high}} = 324$ are both unimodal in the stationary and Binomial prolongations (both peak at the correct fixed point). The TV values measure *inter-pair coupling error* (the product-distribution approximation) rather than prolongation error.

Remark 10.6 (Symmetric IC and the symmetry trap). For the symmetric initial condition $(n_{\text{low}}, n_{\text{low}}, n_{\text{high}}, n_{\text{high}})$, both Binomial and Dynamic- π prolongation give the same coarse-to-fine mapping (both modes are unimodal for $n_c = 2n_{\text{low}}$ and $n_c =$

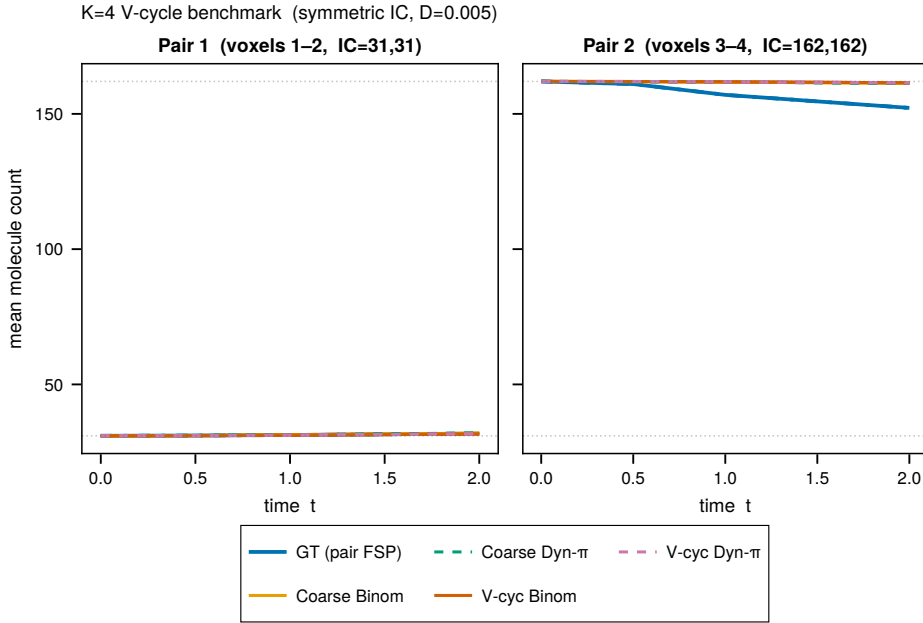


Fig. 5: Per-voxel mean molecule counts vs. time for the $K = 4$ V-cycle benchmark (symmetric IC, $D = 0.005$). All methods track the ground truth (pair FSP, solid blue) closely for the mean. The high TV distances in Figure 6 arise from distributional rather than mean errors.

Table 6: $K=4$ V-cycle benchmark: TV distance vs. pair-level ground truth and wall-clock timing. All methods solve from $t = 0$ to $t_f = 2$ with step size $\Delta t = 0.2$. V-cycle methods use pre-smoothing time $\tau_{\text{pre}} = 1/(dn_{\text{low}} + 1) \approx 0.39$ s.

Method	TV($t=0.5$)	TV($t=1$)	TV($t=2$)	Time (s)	Fine states
Coarse Binom- π	0.954	0.915	0.856	11.2	$\lesssim 10^3$
Coarse Dyn- π	0.954	0.914	0.852	1.7	$\lesssim 10^3$
V-cycle Binom- π	0.977	0.938	0.877	27.9	576,449
V-cycle Dyn- π	0.977	0.938	0.875	5.4	506,229

601 $2n_{\text{high}}$). The symmetry trap only manifests for *asymmetric* coarse states such as
602 $n_c = n_{\text{low}} + n_{\text{high}} = 193$, as demonstrated in Benchmark 2a. For applications where
603 all spatial fronts are at coarse boundaries (i.e. the coarse grid aligns with the front),
604 the symmetric IC benchmark measures the inter-pair coupling error of the V-cycle
605 architecture independently of the prolongation operator.

606 11. Discussion and Outlook.

607 *Dual lumpability as the organizing principle..* The central contribution of this
608 work is recognizing that the RDME Markov chain admits lumpability in two indepen-
609 dent dimensions: the molecular-count dimension (fast reactions $\rightarrow$ QSS aggregation)
610 and the spatial dimension (fast diffusion $\rightarrow$ voxel merging). The companion FP-FSP

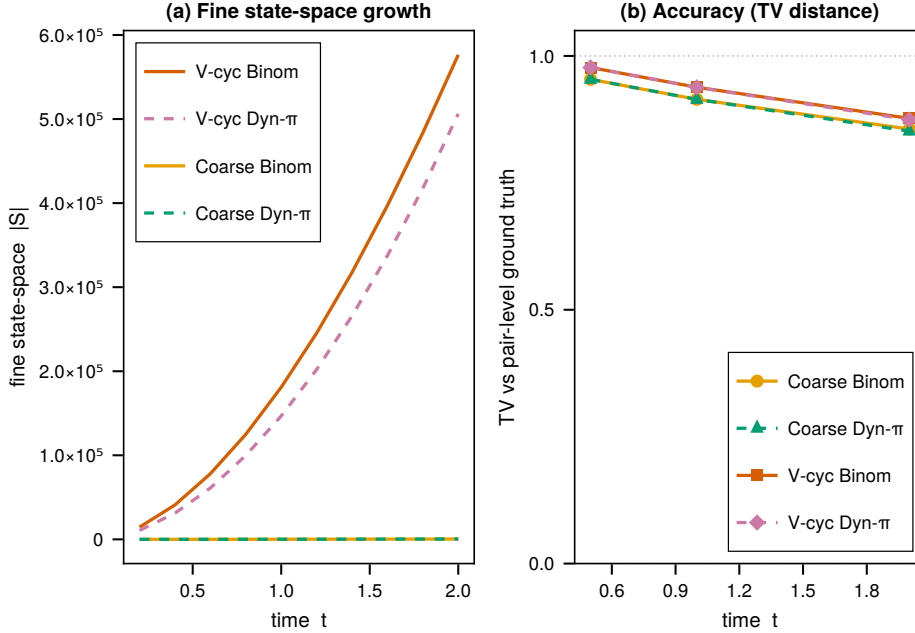


Fig. 6: **(a)** Fine-level state-space size over time. V-cycle methods grow to $\sim 5 \times 10^5$ fine states by $t = 2$; coarse-only methods maintain $\lesssim 10^3$ coarse states. **(b)** TV distance vs. pair-level ground truth. All four methods produce similar TV values (0.85–0.97) because the error is dominated by inter-pair coupling (the independence assumption in the proxy GT) rather than prolongation error. Dynamic- π V-cycle runs $5\times$ faster than Binomial V-cycle (5.4 s vs. 27.9 s) with a smaller fine state space (506k vs. 576k states).

work [4] operates in the molecular-count dimension; the present multigrid method operates in the spatial dimension. Both coarsenings are instances of the same mathematical operation (Definition 3.1), driven by the same lumpability parameter (ratio of inter-group to intra-group transition rates), and require the same bottleneck protection (Section 3.4).

A fully adaptive RDME solver would perform both coarsenings simultaneously: aggregating fast-reaction chains in each voxel (reducing the per-voxel state space) while merging slow-diffusion voxel pairs (reducing the number of voxels). The resulting two-dimensional coarsening hierarchy acts on the full product state space $\mathbb{Z}_{\geq 0}^{N \times K}$, with each dimension coarsened independently according to its own lumpability parameter.

Relation to QSS methods. Quasi-steady-state (QSS) approximations for the CME [12] project out fast species by assuming they reach their conditional stationary distribution instantaneously. Our multigrid framework recovers this limit when $L = 1$ and $\tau_{\text{pre}} \rightarrow \infty$: the pre-smoothing step equilibrates the intra-voxel distribution, and the coarse solve handles only the slow inter-voxel dynamics. From the lumpability perspective (Section 3), QSS is approximate strong lumpability in the molecular-count

dimension, while our voxel merging is approximate strong lumpability in the spatial dimension. The key difference from classical QSS is that we quantify the approximation error via ϵ_{smooth} and $\epsilon_{kk'}^{(\ell)}$ (Theorem 7.1 and Theorem 3.3), whereas classical QSS has $O(\epsilon)$ errors without sharp bounds.

Tensor-train methods.. Tensor-train (TT) decompositions [10] exploit the same tensor-product structure (2.6) to compress the probability tensor. TT-cross approximation of the RDME distribution is an alternative computational approach. The multigrid method complements TT methods: while TT reduces storage by exploiting low-rank structure, multigrid reduces the effective time-stepping stiffness by hierarchical level decomposition. A combined multigrid-TT method is a natural direction for future work.

RDME vs. SSME.. At very fine spatial resolution (voxel size comparable to the molecular interaction radius), the RDME becomes inconsistent and must be replaced by the *spatially continuous master equation* (SCME) or related formulations [9]. The multigrid framework developed here applies at the RDME level of description; extending it to continuous-space models is an open problem.

Higher-order prolongation.. The binomial prolongation (4.7) uses only the total count $\bar{x}_{ij}$ to reconstruct the within-voxel split. A higher-order prolongation could incorporate correlations from the preceding fine-level distribution, analogous to the *injection* vs. *interpolation* prolongation operators in classical geometric multigrid.

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